Scientific Computing

Mathematical Foundations for Machine Learning

Seminar 1: Problem Set

Probability Models, Conditioning, and Random Variables

Block I, following Lecture 1

2026

Problem 1 (The Monty Hall protocol). Three doors. Behind one of them is an internship in a decent laboratory; behind the other two, goats. The prize door is uniform over the three doors, independently of your initial choice. You point at a door. The host, who knows what is behind each door, opens an unchosen door hiding a goat and offers you the chance to switch. If both unchosen doors hide goats, he chooses between them with equal probabilities. Write c for the prize door, g for your initial guess, and h for the door the host opened.

(a) Construct a probabilistic model of the experiment.

(b) Assume you guessed door 1. Compute $\mathbb{P}(c = 1 \mid h = 3)$ and $\mathbb{P}(c = 2 \mid h = 3)$ directly from the definition of conditional probability. Should you switch?

(c) Identify the exact place in your computation where the assumption that the host knows the answer is used.

Remark. This is the question put to a student in the film *21* (2008); answering it correctly earns him a seat on a card-counting team. Evidence that a conditional probability exercise occasionally changes a life.

Problem 2 (Monty Fall). The same three doors, but this time the host does not know what is behind them. He trips, and in falling pushes open one of the two doors you did not choose, each with probability 1/2, independently of the prize. It happens to reveal a goat. He offers you the chance to switch.

(a) Compute the probability that the prize is behind your original door, given that a goat was revealed. Should you switch?

(b) You see exactly what you saw in Problem 1: your door closed, one other door open, a goat behind it. Explain why the answer is nevertheless different.

(c) Which single quantity in the model of Problem 1 has changed?

Problem 3 (A detector with 95% sensitivity). Consider a university that runs submitted essays through an AI-text detector. Assume the detector has the following class-conditional probabilities:

$$\mathbb{P}(\text{flag} \mid \text{AI}) = 0.95, \qquad \mathbb{P}(\text{flag} \mid \text{human}) = 0.05.$$

Assume one essay in twenty is AI-generated, so $\mathbb{P}(\text{AI}) = 0.05$. Your essay has been flagged.

(a) Compute $\mathbb{P}(\text{AI} \mid \text{flag})$.

(b) How large would the base rate $\mathbb{P}(\text{AI})$ have to be for a single flag to give a posterior probability above 0.9?

Problem 4 (Two detectors, and the odds form of Bayes' rule). The university adds a second detector with the same two characteristics, and both detectors flag your essay. Write H for the hypothesis "AI-generated", H^c for its complement, and $E_1, \dots, E_n$ for the events that detectors $1, \dots, n$ raise a flag. Assume that the flags are mutually conditionally independent given each class.

(a) Compute $\mathbb{P}(H \mid E_1 \cap E_2)$ directly from Bayes' theorem.

(b) Define the posterior odds and prior odds

$$\mathcal{O}(H \mid E) = \frac{\mathbb{P}(H \mid E)}{\mathbb{P}(H^c \mid E)}, \qquad \mathcal{O}(H) = \frac{\mathbb{P}(H)}{\mathbb{P}(H^c)},$$

and the likelihood ratio of a single flag, $\mathrm{LR} = \mathbb{P}(E_i \mid H)/\mathbb{P}(E_i \mid H^c)$. Write Bayes' theorem for H and H^c, take the ratio, and cancel the common normalizing denominator. Using conditional independence in both classes, deduce that for n detectors with identical characteristics

$$\mathcal{O}(H \mid E_1 \cap \cdots \cap E_n) = \mathrm{LR}^n \, \mathcal{O}(H).$$

Convert back with $P = \mathcal{O}/(1+\mathcal{O})$, check that $n = 2$ reproduces part (a), and find the smallest n for which the posterior probability exceeds 0.99.

Remark (Dependent flags). Without conditional independence, replace LR^n by the actual joint likelihood ratio

$$\frac{\mathbb{P}(E_1 \cap \cdots \cap E_n \mid H)}{\mathbb{P}(E_1 \cap \cdots \cap E_n \mid H^c)}.$$

Dependence alone does not tell us whether this ratio is above or below the product of the individual ratios: the joint probabilities in both classes determine it. If the second detector always copies the first, then $E_2 = E_1$, and two flags give exactly the same posterior as one. Multiplying the two likelihood ratios would count the same evidence twice.

Problem 5 (The best of several runs). A random search launches two independent training runs. Their qualities R_1, R_2 are uniform on $\{1, 2, \ldots, 6\}$ on a fixed scale. You keep the better of the two: $X = \max(R_1, R_2)$.

(a) Find the CDF $F_X(k)$ for $k = 1, \ldots, 6$ before attempting the PMF, and explain why the CDF is easier to calculate here.

(b) Recover the PMF using $p_X(k) = F_X(k) - F_X(k^-)$.

(c) Compute $\mathbb{P}(X \geq 5)$.

(d) Generalize to n independent runs. How large must n be for $\mathbb{P}(X = 6) > 0.9$?

Problem 6 (Densities and inverse-CDF sampling).

(a) Let $f_Y(y) = 10$ for $y \in [0, 0.1]$, and zero otherwise. Verify that f_Y is a density and compute $\mathbb{P}(Y = 0.05)$.

(b) Let $f_X(x) = 2x$ on $[0, 1]$, and zero elsewhere. Find F_X and its inverse F_X^{-1} on $[0, 1]$.

(c) Let $U \sim \mathrm{Uniform}(0, 1)$. Show that $X = F_X^{-1}(U)$ has CDF F_X. Begin with $\mathbb{P}(X \leq x)$.

(d) Write down the same construction for the exponential distribution with $F(x) = 1 - e^{-\lambda x}$, $x \geq 0$.